

Received December 13, 2021, accepted January 15, 2022, date of publication January 21, 2022, date of current version January 27, 2022.

Digital Object Identifier 10.1109/ACCESS.2022.3145426

Three-Way Ensemble Prediction for Workload in the Data Center

RUI SHI¹ AND CHUNMAO JIANG¹

School of Computer Science and Information Engineering, Harbin Normal University, Harbin, Heilongjiang 150025, China

Corresponding author: Chunmao Jiang (chunmaojiang@hrbnu.edu.cn)

This work was supported in part by the Natural Science Foundation of Heilongjiang Provincial under Grant LH2020F031, and in part by the Postgraduate Innovation Project of Harbin Normal University under Grant HSDSSCX2021-30.

ABSTRACT Accurate prediction of data center workload, an important technology of cloud computing, is particular in improving resource utilization and reducing energy consumption. However, the workload presents a quasi-volatile, is challenging to obtain accurate results in cloud resource management. In this paper, the three-way ensemble prediction for workload in the data center is first proposed to improve the accuracy of the prediction. Moreover, we first defined the workload as the stable period, the volatility period, and the jitter period and adopted a simulated annealing algorithm to learn the optimal threshold to divide the workload. Then, according to the basic idea of the three-way decision, we assigned various prediction models based on workload characteristics and a priori error prediction to improve the prediction accuracy further. Finally, all the experimental results carried on the CPU load monitoring logs from Google cluster trace and compared with ARIMA, NN, and DMASVR-3WD, TWD-RCPM improve the accuracy of workload prediction by 69.0%, 68.6% and 72.6%, respectively.

INDEX TERMS Three-way decision, three-way division, cost assessment, load prediction, cloud energy consumption.

I. INTRODUCTION

Recently, the rapid expansion of cloud data center scale, which is bound to aggravate the continuous rise in energy consumption [1-4]. The cause of energy consumption may be a variety of factors such as data center resource demand, refrigeration equipment, and network equipment [5]. Among them, data center resource demand is a crucial factor affecting energy consumption: 1) when the resource supply is too high, the host will be idle and the resource utilization will decrease; 2) when the resource supply is too low, the host will increase the task wait time resulting in higher Service level agreement (SLA) violation. Both cases will increase data center energy consumption [6-7]. Therefore, improving the prediction accuracy of data center resource demand will better confirm the supply of resources, to improve resource utilization and decrease energy consumption. In short, it is important to determine the accurate number of resources in advance.

To achieve the balance of resource demand and resource supply, the prediction approaches play a critical key to

determine the number of resources. One of the most challenging tasks is how to improve prediction accuracy [8]. A wide range of prediction modeling techniques has been used for estimating the expected resource demands in the data center. The prediction methods are mainly based on time series models such as moving average model (MA) [9], autoregressive moving integrated average model (ARIMA) [10], support vector machines (SVM) [11], neural network (NN) [12], long short-term memory network (LSTM) [13]. However, with the expansion of the data center scale, it is a new challenge to workload prediction in the data center in the diversity of resource demand characteristics and the heterogeneity of resource type [14]. The above models that most of the predictive frameworks use a single model to anticipate future workload and their accuracy tends to drop down as the pattern of workloads changes. Apart from the above-mentioned methods, several resource prediction work in the cloud data center has made some progress around the characteristic mode of the workload. For example, there are certain rules between specific task types and task resource demand, Sun *et al.* [15] predicted the host workload by combining support vector machine (SVM) and backpropagation neural network (BP) through mining

The associate editor coordinating the review of this manuscript and approving it for publication was Ahmed Diab¹.

hidden rules. Patel *et al.* [16] focused on the problem of identifying and predicting patterns in cloud workloads, and they proposed a clustering approach based on resource usage to identify periodic tasks and non-periodic tasks. Babu *et al.* [17] divided the workload into two types: linear and nonlinear, and applied ARIMA-ANN combined model for classification prediction. Calzarossa *et al.* [18] considered the periodic and burstiness of workload, proposed a model based on trigonometric polynomials and ARMA to dynamically reproduce the data change pattern, to provide additional data for workload prediction. However, the above techniques can only predict workload by data feature and cannot capture the workload fluctuation. Therefore, this paper carried out modeling for workload fluctuation, and proposed a data center workload prediction method based on the three-way decision, to improve prediction accuracy. We summarized the related work into the following Table 1.

The main contributions of this paper are as follows: 1) Based on the diversity of workload fluctuation in a large-scale heterogeneity cloud, we proposed an adaptive classified prediction scheme. By analyzing the historical workload fluctuation, we reconstructed the risk loss function. This paper used the simulated annealing algorithm to learn the optimal thresholds. 2) To obtain a joint prediction of multiple load sequences, we applied targeted strategies to the workload in different types. 3) Finally, we used LSTM to make a priori error prediction to correct the joint prediction results.

The rest of this paper is organized as follows: Section II, reviews the basic prediction model, long short-term memory network and the three-way decision. Section III, proposed the three-way ensemble prediction method (TWD-RCPM) for data center workload. Section IV, gives the experimental results and analysis. Finally, the conclusion is presented in Section V.

II. PRELIMINARIES

We adopted CPU load monitoring logs from the Google traces. In a visual observation of the change, Figure 1 shows three examples of workload change patterns, namely, the stable period, the volatility period, and the jitter period.

Data (a) varies steadily; data (b) has an obvious fluctuation trend; data (c) shows a sharp climb or decline with a high degree of dispersion between points. Effective information extraction and utilization of time series data is important for improving prediction accuracy.

This section briefly reviews some basic method related to: 1) the basic prediction models; 2) long short-term Memory networks; 3) three-way decision.

A. STABLE PERIOD PREDICTION MODEL

The workload in the stable period has less fluctuation and low dispersion, we adopted an Exponential Moving Average model (EMA)[19]. The model ensemble recent data to predict the workload at the next time. The model has a time interval that is evenly divided into windows. Given a sliding window

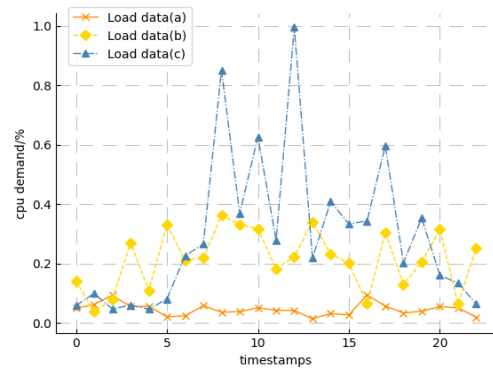


FIGURE 1. CPU demand load characteristics.

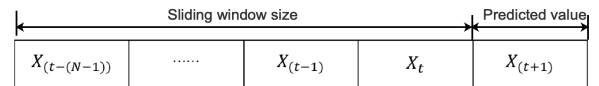


FIGURE 2. The model of sliding window.

$X = [X_{t-(N-1)}, X_{t-(N-2)}, \dots, X_t]$ with length N , the time interval is $(t - (N - 1), t]$. They are detailed in Figure 2.

Consider the workload sequence as a set of dependent variables $X_t, t \in \{1, 2, \dots, N\}$, X_t stands for the actual workload in window t . Then the predicted value X_{t+1} can be expressed as Equation (1).

$$X_{t+1} = f(X, N) \quad (1)$$

where X_{t+1} is the predicted value of workload on the time series, and $t + 1$ is the time series label. The model uses factor α to weight the workload at N time windows, and the calculation formula of X_{t+1} can be written as Equation (2):

$$X_{t+1} = \alpha \left[P(t) + (1 - \alpha)P(t - 1) + (1 - \alpha)^2 P(t - 2) + \dots + (1 - \alpha)^{N-1}P(1) \right] \quad (2)$$

Equation (2) can be written in the recursive form:

$$X_{t+1} = \alpha P(t) + (1 - \alpha)X_t \quad (3)$$

According to above analysis, the predicted value X_{t+1} is determined by the window length of the workload sequence. The model has a good ability to process steady-state data, which makes the predicted value more gradual by this method is gentler and smoother. The predicted value will not deviate from the moving average due to some outlier of historical data.

B. JITTER PERIOD PREDICTION MODEL

The data in the jitter period has a large fluctuation and high dispersion. When the time series shows a conic curve trend, we adopted the Holt-Winters model [20]. Holt-Winters is a statistical model, also known as the triple exponential smoothing model. This model is based on first exponential smoothing and second exponential smoothing. The first

TABLE 1. Summary of related work.

Literature	Research theme	Key finding	Proposal
G. Chen et al. [9]	moving average (MA)	An approach for long-term trend prediction using MA method	Predict data trends using only a single model
R. N. Calheiros et al. [10]	auto-regressive moving integrated average (ARIMA)	An approach for the realization of a Cloud workload prediction module based on the ARIMA	Predict resource demand using only a single model
N. I. Sapankevych et al. [11]	support vector machines (SVM)	A survey of time series prediction using SVM	Using SVM is the ability to accurately predict time series when the data are non-stationary
Y. C. Chang et al. [12]	neural network (NN)	The NN model with learning algorithm to predict the workload of the cloud	Accuracy tends to drop down as the pattern of workload changes
H. M. Nguyen et al. [13]	long short-term Memory network (LSTM)	A prediction approach based on LSTM Encoder-Decoder (LSTM-ED)	The model has low accuracy when the workload data changes sharply
C. H. Liu et al. [14]	SVM and linear regression(LR)	An approach classified workload of the workload into fast time-scale or slow time-scale	Misclassification cost is not considered for assignment of task type
Q. Sun et al. [15]	SVM and back-propagation (BP)	An approach with SVM and BP to predict the workload of host	Unable to capture workload characteristics
J. Patel et al. [16]	clustering and integrated computing	A cluster-based prediction approach that groups tasks with similar features into the same cluster	The intensity of the task workload change is not considered
C. N. Babu et al. [17]	ARIMA and artificial neural network (ANN)	A hybrid ARIMA-ANN model for the prediction of time series data	Only the linear and non-linear relationship of the data is considered
M. C. Calzarossa et al. [18]	trigonometric polynomials and ARMA	An approach predict the empirical change patterns and provide data into the future for forecasting	Only trend and seasonal of the workload are considered, volatility cannot be captured

exponential smoothing method is used for time series with no clear law but some correlation. The second exponential smoothing preserves the trend information of the time series. Let the workload sequence $X = [X_1, X_2, \dots, X_{t-1}]$, then the predicted value X_t can be expressed as Equation (4).

$$\begin{cases} S_t^{(1)} = \alpha X_{t-1} + (1 - \alpha)S_{t-1}^{(1)} \\ S_t^{(2)} = \alpha S_t^{(1)} + (1 - \alpha)S_{t-1}^{(2)} \\ S_t^{(3)} = \alpha S_t^{(2)} + (1 - \alpha)S_{t-1}^{(3)} \end{cases} \quad (4)$$

where α is the weighting coefficient ($0 < \alpha < 1$), which controls the influence of the historical workload on the current predicted time. Based on experience, the value of α should be set in the range of 0.1 to 0.8 [21]. X_{t-1} is the actual workload at time $t - 1$, $S_t^{(1)}$, $S_t^{(2)}$ and $S_t^{(3)}$ is first smoothing value, second smoothing value, and predicted value of Holt-Winters.

C. LONG SHORT-TERM MEMORY

The hidden layer is the core of time series information extraction [22]. By treating the hidden layer as a memory unit, the LSTM network can cope with the correlation within time series in both the short and long term. There is gate mechanism in the memory unit, which is composed by forget gate f_t , input gate i_t , and output gate o_t [23]. Figure 3 shows the structure of the memory unit:

At time t , h_{t-1} represents the output state of the last neural unit, X_t is the input state of the current memory unit, and W is the learnable weight matrix. Firstly, some information is abandoned through the forget gate f_t . For example, the memory unit will forget the outliers in the historical data and

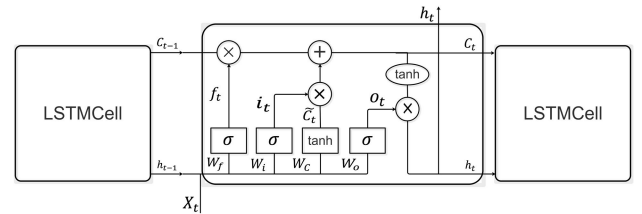


FIGURE 3. LSTM neural unit basic structure diagram.

relatively old data with the current time.

$$f_t = \sigma(W_f \cdot [h_{t-1}, X_t] + b_f) \quad (5)$$

Secondly, the input gate i_t decides what new information to keep in the current memory unit. For example, there are some relations and rules between time series, i_t uses the activation function to map the filtered data to (0, 1), meanwhile, the memory unit using the $\tanh$ layer to create a new vector, which represents the weight of data updated to the LSTM memory unit.

$$i_t = \sigma(W_i \cdot [h_{t-1}, X_t] + b_i) \quad (6)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, X_t] + b_C) \quad (7)$$

Then the memory unit updates the old state C_{t-1} to the new state C_t .

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (8)$$

Finally, the output gate o_t calculates the actual output.

$$o_t = \sigma(W_o \cdot [h_{t-1}, X_t] + b_o) \quad (9)$$

Via the function of the gate mechanism, LSTM memory units can capture the complex correlation features within time

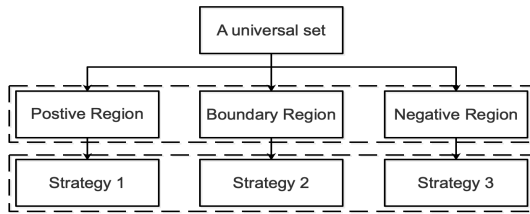


FIGURE 4. Three-way decision model.

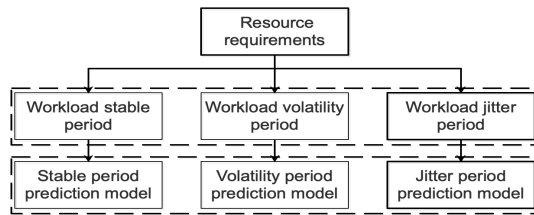


FIGURE 5. Load prediction model.

series in both the short and long term. The network carries out data memory and data forgetting to realize long-term memory and solve the long-term dependence problem of sequential data [24].

D. THREE-WAY DECISION

By extracting ideas of thinking in threes from many different fields, Yao introduced the three-way decision [25]. The theoretical framework is shown in Figure 4. Three-way decision is the information processing ability based on human cognition, which is more efficient decision-making and information processing [26]. Thinking in threes is a core idea of the three-way decision. The three fundamental issues of the three-way decision are how to construct a trisection, how to develop action and strategy on the three regions [27].

With the deepening of cloud computing theory research, the utility of the three-way decision has become increasingly prominent. There are many tripartite phenomena in cloud computing architecture. For example, according to resource usage, the VMs could be divided into idle state nodes, the normal nodes and overload nodes. Jiang *et al.* [28] proposed an adaptive thresholds determination for saving cloud energy using three-way decisions. The mobile cloud task offloading method could be divided into immediate offloading, delayed offloading, and no offloading. Xu *et al.* [29] proposed an energy-saving offloading strategy for mobile cloud tasks oriented to tolerance delay.

The three-way decision divides a universal set into three pair-wise disjoint regions and applies targeted strategies to different regions [30-32]. The method transforms a complex problem into simple problems. The basic principle could just make up for the deficiencies of previous studies and fit the problem that this paper needs to solve. In this paper, we proposed the three-way ensemble prediction for workload in the data center. We defined workload as three different

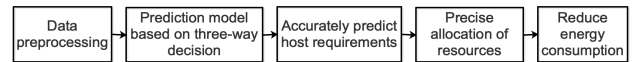


FIGURE 6. Flow chart of energy efficiency optimization.

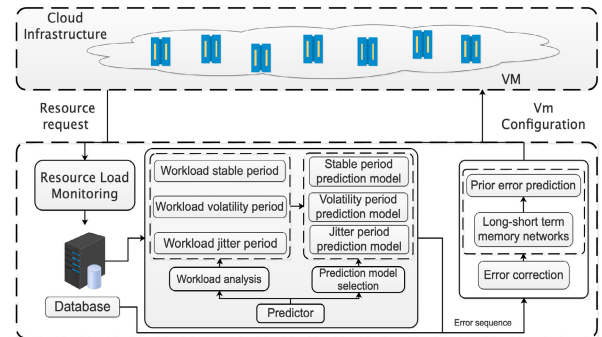


FIGURE 7. Workload prediction architecture diagram.

types and applied prediction models in a targeted manner. The application framework is shown in Figure 5.

III. DATA CENTER WORKLOAD PREDICTION MODEL

The model enables the data center to predict the resource demands. This process can dynamically adjust resource allocation to improve resource utilization and reduce power consumption. The complete process is shown in Figure 6.

A. PREDICTION MODEL ARCHITECTURE

The overall proposed architecture for our model is shown in Figure 7 that consists four modules including Resource Load Monitoring module, Database module, Predictor module, and Error Correction module. VM refers to the virtual machine.

The Resource Load Monitoring module plays the role of storing and providing monitoring data. There are two data types: historical and real-time. The first type is data used for the preprocessing and training model. The second data type is workload utilization gathered in real-time and put to the trained model to predict resource demand in advance. The real-time data will be stored in the historical data for the process of updating the prediction model. These data are sent to the database. In preprocessing stage, the historical data first is used to analyze the workload characteristics and get the threshold for dividing data, and then which used to train the LSTM prediction model. The Predictor module. In the Predictor module, the resource workload is a multivariate time series, and the sliding window technology is applied to simulate online prediction. According to the threshold to determine the period of the current workload, and then assign corresponding prediction models. Based on the Error Correction module, we could get the final prediction result, which makes the data center to estimates the number of VMs. The data center compares the expected number of VMs with the number of provisioned VMs, and dynamically adjusts the number of allocated VMs to implement the horizontal or vertical scaling on resources and improve resource utilization.

TABLE 2. Load prediction cost function.

	a_P	a_B	a_N
X	λ_{PP}	λ_{BP}	λ_{NP}
X^C	λ_{PN}	λ_{BN}	λ_{NN}

The system has a scalability. For example, data center energy consumption is mainly related to CPU utilization, but not linearly related to CPU utilization [33]. We can consider memory requirements in the predict process, and we can also introduce more prediction models to expand the system.

B. THREE-WAY DIVISION BASED ON COST ASSESSMENT

The notion of three-way decision rules is introduced based on three disjoint regions produced by the lower and upper approximations [34]. Given a state set $\Omega = \{X, X^C\}$ represents the workload types, where X denotes the stable period, X^C denotes the jitter period. Given an action set $D = \{a_P, a_B, a_N\}$ represents three decision-making actions, namely decide $x \in \text{POS}(X)$, $x \in \text{BND}(X)$, and $x \in \text{NEG}(X)$. Different actions generated different costs, the loss function matrix in the different states is given in Table 2.

In the matrix, λ_{PP} , λ_{BP} , and λ_{NP} denote the losses incurred for taking actions a_P , a_B , and a_N , respectively, when an object belongs to X , λ_{PN} , λ_{BN} , and λ_{NN} denote the losses incurred for taking the same actions when the object belongs to X^C . We adopted the Mean Square Error (MSE) to measure the cost of different actions. The mathematical representation of the metric is mentioned as follows.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \omega_i (y_i - \hat{y}_1)^2 \quad (10)$$

C. THREE-WAY DECISION THRESHOLD DETERMINATION BASED ON SIMULATED ANNEALING

In the binary classification problem, where $\alpha + \beta = 1$, the actions a_P , a_B , a_N are only relative to X , X^C , and the loss function λ . We calculated the window volatility, noted as p , and compared with (α, β) . For $p \geq \alpha$, we adopted positive rule; for $p < \beta$, we adopted negative rule and for $\beta < p < \alpha$, we adopted boundary rule. The threshold parameters α and β are derived from the six given loss functions. We assumed $\lambda_{NN} = \lambda_{PP} = 0$ which means correct classification does not incur any cost. We assumed $\lambda_{BP} = \lambda_{BN}$ which means the cost of misclassification of the stable period and the jitter period into the volatility period is the same. Therefore, α and β are only related to the remaining four cost functions. The total cost is calculated as follows.

$$\begin{aligned} \text{cost}_A = & \sum_{x \in \text{POS}_{(\alpha, \beta)}(\pi_D | \pi_A)} \lambda_{PN} \\ & + \sum_{x \in \text{BND}_{(\alpha, \beta)}(\pi_D | \pi_A)} \lambda_{BP} \\ & + \sum_{x \in \text{NEG}_{(\alpha, \beta)}(\pi_D | \pi_A)} \lambda_{NP} \end{aligned} \quad (11)$$

The optimization problem with minimizing loss can be expressed as:

$$\text{mincost}_A = \sum_{p \geq \alpha} \lambda_{PN} + \sum_{\beta \leq p < \alpha} \lambda_{BP} + \sum_{p < \beta} \lambda_{NP} \quad (12)$$

Based on the discussion above, we can obtain the solution to the optimization problem [35]. The simulated annealing (SA) acceptance of “bad” solutions in order to escape from local optimal configurations. In load prediction problem, a set of parameters can be prevented from entering local optimum due to the accuracy of certain prediction. We set the solution space as samples probability. Given a random initial solution, at each iteration of the simulated annealing, a new neighbor x^* is taken from the current solution neighborhood $N(x)$. If the move to the neighbor decreases the objective function value $f(x)$ or leaves it unchanged, then the move is always accepted. Thus, the solution x^* becomes the new current solution if $dE < 0$, where $dE = f(x^*) - f(x)$; otherwise, the solution x^* is adopted as the new current solution with a certain probability. Repeat the process until the temperature limit is reached [36]. The algorithm 1 is as follows.

Algorithm 1: Simulated Annealing Algorithm

Input: workload sequence; sample probabilities; loss cost;
Output: threshold (α, β) ;
Initialize the parameters of the annealing schedule;
Randomly generate an initial state as current solution x ;
while $T_i > T_{\min}$ **AND** the result is not convergent **do**
 Generate the neighbor solution $f(x)$ based on the current solution x^* ;
 Compute the fitness function $f(x)$ and $f(x^*)$;
 // $f(x)$: Indicates current status;
 // $f(x^*)$: Indicates a new state;
 $dE = J(f(x^*)) - J(f(x))$;
 if $dE \geq 0$ **then**
 Accept the move from x to x^* ;
 else
 $p = \exp(dE/T)$;
 if $\text{random.rand} < p$ **then**
 Accept the move from x to x^* ;
 $T = r \cdot T$;
return (α, β) ;

where $f = \text{cost}_{(\alpha, \beta)}$ is the objective function; T as the initial temperature, set to 2000. At the end of an iteration, gradual attenuation of T_{i+1} is expressed as $T_{i+1} = r \cdot T_i$ where r is the temperature cooling parameter [34]. ρ is a random number between 0 to 1 which control x^* can be accepted as the current solution in a certain probability. The time complexity of the algorithm 1 is $O(n)$, and the space complexity is $O(1)$.

In order to improve the prediction accuracy, we adopted the Root Mean Squared Error (RMSE) to calculate the

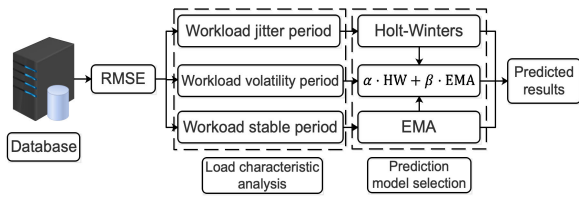


FIGURE 8. System architecture of three-way decisions model.

fluctuation value of the time series.

$$h_t = \sqrt{\frac{1}{L} \sum_{i=t-(\omega-1)}^t (x_{i+1} - x_i)^2} \quad (13)$$

The RMSE between adjacent moments in the workload is noted as h_t . Compared with threshold, $h_t > \alpha$ is considered that the workload changes drastically; $h_t < \beta$ is considered that the workload changes smoothly. Given a historical workload X_t , $t \in \{1, 2, \dots, N\}$, $h_t^{(i)} \geq \beta$ we considered that the resource is in the jitter period; $h_t^{(i)} \leq \alpha$ we considered that the resource is in the stable period; $\beta \leq h_t^{(i)} \leq \alpha$ we considered that the resource is in the fluctuation period.

D. PREDICTION MODEL BASED ON THREE-WAY DECISION

An ensemble approach involves the use of multiple prediction models to predict the estimated future outcome and each of them is commonly referred to as a base predictor. With the help of a three-way decision, by analyzing the workload in Google Cluster Trace 2011, we divide the workload into a positive domain (POS), negative domain (NEG), and boundary domain (BND). The conceptual architecture of an ensemble-based predictive approach is depicted in Figure 8.

For the stable period, we adopted the EMA prediction model. For the jitter period, we adopted the Holt-Winters prediction model capable of respond to rapid changes in data. For the volatility period, we adopted a weighted prediction method. At time t , the outcome of the basic prediction model i ($i = 1, 2$) is $h_t^{(i)}$, and the corresponding weight is $\omega_t^{(i)}$ then the prediction result of the volatility period is formalized as follows.

$$h_t = \sum \omega_t^{(i)} h_t^{(i)}; \quad \text{s.t.} \quad \sum \omega_t^{(i)} = 1 \quad (14)$$

The base prediction model of the predictive framework produces their corresponding outcome and each of them is weighted to compute the final predicted workload. We use the reciprocal error method [38] to calculate the weight to ensure that the model with high accuracy has a large weight, and the model with low accuracy has a small weight. The calculation method is as follows:

$$\omega_t^{(i)} = \frac{e_t^{(i)}}{\sum_i e_t^{(i)}} \quad (15)$$

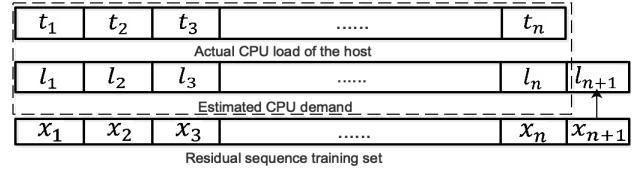


FIGURE 9. Error correction.

E. ERROR CORRECTION METHOD

As is known to all, there will be errors in the prediction model, and error correction is a general method to solve this problem [39]. The error correction process is shown in Figure 9.

Given $T_i = \{t_1, t_2, \dots, t_n\}$ is the actual CPU load, $L_i = \{l_1, l_2, \dots, l_n\}$ is the ensemble prediction outcome, then we obtained the error sequence $X_i = L_i - T_i$ ($i = 1, 2, \dots, n$). We adopted X_i to train the LSTM model.

At time $n + 1$, the predicted CPU load is l_{n+1} . Meanwhile, we adopted LSTM to obtain the error predicted value x_{n+1} used for error correction. Then we obtained the final predicted value $Final_p = l_{n+1} + x_{n+1}$. Based on the discussion above, the algorithm is as follows. The time complexity of the algorithm 2 is $O(n)$, and the space complexity is $O(1)$.

Algorithm 2: TWD-RCPM Prediction Model

Input: T_i : A data sequence of the host's historical load
 $T_i = \{t_1, t_2, \dots, t_n\}$;
Output: $Final_p$: Predicted value of CPU requests;
 $h_t = \sqrt{\frac{1}{L} \sum_{i=t-(n-1)}^t (x_{i+1} - x_i)^2}$;
for $i = N; i < m; i + +$ **do**
 // m is the length of the time series;
 // N is the size of the window;
 $E_{pred} = \text{exp_moving_average}(T_i, \alpha)$;
 $HW_{pred} = \text{Holt-Winters}(T_i, \alpha)$;
 $L_{pred} = \text{cal}(E_{pred}, HW_{pred})$;
if $h_t^{(i)} \leq \alpha$ **then**
 $l_{n+1} = E_{pred}$;
else if $h_t^{(i)} \geq \beta$ **then**
 $l_{n+1} = HW_{pred}$;
else
 $l_{n+1} = L_{pred}$;
for $i = 0; i < n; i + +$ **do**
 $X_i = L_i - T_i$;
 $x_{n+1} = \text{LSTM}(X_i)$;
 $Final_p = l_{n+1} + x_{n+1}$;
return $Final_p$;

IV. EXPERIMENTAL ANALYSIS

A. EXPERIMENTAL ENVIRONMENT

The experiment is a computer equipped with 64-bit macOS Mojave operating system, 8GB RAM, and 256G hard disk capacity. The experiment is coded in python. We conducted

experiments by using real-workload trace sampled from Google data center, we counted the CPU load of the entire data center. In this paper, we model the CPU load fluctuations of the data center. The first 80% data is used as the training set, and the last 20% data is used as the test set to verify the generalization ability of the model. To simulate the online prediction, the remaining data points are gradually added to the historical data set for the next prediction.

B. EXPERIMENT PREPARATION

Various services are running in data centers, and the heterogeneity of resource demand characteristics and diversity of resource types are the problems that must be considered in load prediction. We adopted CPU monitoring logs from the Google traces. The cluster machines are heterogeneous; they consist of three different platforms. Exact numbers of CPU cores are unavailable; instead, CPU size measurements are normalized to the configuration of the largest machines. The cluster scheduler assigns tasks to machines. Data center applications fall into four types: CPU intensive, I/O intensive, memory intensive, and network intensive, which consume CPU, I/O, memory, and network bandwidth in sequence. Energy consumption is mainly affected by CPU utilization [40]. The cluster scheduler is done in order by task type, first-come, first-served for tasks of the same type. The “trace” consists of a month of activity in a single 12000 machine heterogeneous cluster. Reiss *et al.* [41] measures resource usage of the Google dataset, averaging over one-hour windows, CPU usage does not exceed about 60%. The trace describes hundreds of thousands of jobs, including per-task resource usage. The dataset was sampled in a five-minute cycle. To compute the actual utilization, for each task usage record available, we took the sum of the average CPU weighted by the length of the measurement.

In this paper, we verified the performance and generalization ability of the TWD-RCPM prediction model by four metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Symmetric Mean Absolute Percentage Error (MAPE), and Qualified Rate (QR).

(1) Accuracy

Accuracy is one of the significant metrics to verify the prediction model performance, and could provide guidance for data center resource scheduling. The accuracy rate is measured by the following three metrics. The mathematical representation of the metric is mentioned as follows.

Mean Absolute Error:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

Root Mean Square Error:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (17)$$

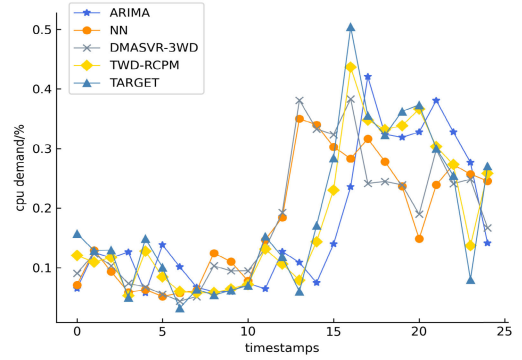


FIGURE 10. Task_usage_499 comparison of experimental results.

Symmetric Mean Absolute Percentage Error:

$$SMAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{(|\hat{y}_i| + |y_i|)/2} \right| \quad (18)$$

(2) Qualified Rate

In cloud resource management, if the resource supply is too high to decrease resource utilization, and the resource supply is too low leading to a higher SLA. Therefore, in order to improve the service quality, we adopted the QR to measure user satisfaction. The mathematical representation of the metric is mentioned as follows.

$$QR = \text{count}(\tilde{y}_i \geq y_i) / \text{count}(y_i) \quad (19)$$

where y_i is the actual workload, and $\tilde{y}_i$ is the predicted workload.

C. COMPARATIVE ANALYSIS OF PREDICTION RESULTS

We compared the prediction effect of TWD-RCPM model with DMASVR-3WD, ARIMA, and NN prediction models respectively. The comparison outcome is shown in Figure 10.

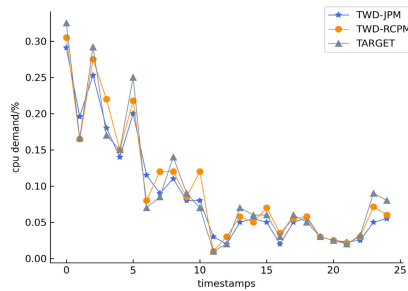
We selected 25 points uniformly within the prediction period, computed the CPU load of the 25 points with four prediction models. In the stable area from point 6 to point 10, ARIMA and TWD-RCPM prediction results are more accurate, but NN and DMASVR-3WD prediction results had larger errors. In the volatility areas from point 0 to point 5 and point 16 to point 22, the ARIMA prediction result had a lag, and the NN and DMASVR-3WD cannot adapt to volatility in data. In the jitter areas from point 11 to point 15 and point 23 to point 25, CPU load had increases by more than 50%. ARIMA, NN, and DMASVR-3WD models cannot adapt to the drastic changes in the CPU load. The TWD-RCPM model can effectively adapt to CPU load changes. Based on the same data set, the TWD-RCPM model had a better fitting effect than the other three models.

The evaluation index calculation results among the four models are shown in Table 3.

In Table 3, the RMSE of ARIMA, NN, and DMASVR-3WD are 0.084, 0.083, and 0.095, respectively, which are all higher

TABLE 3. Comparison of evaluation indexes in Task_usage_499.

	RMSE	MAE	SMAPE	QR
ARIMA	0.084	0.051	0.214	0.55
NN	0.083	0.050	0.215	0.48
DMASVR-3WD	0.095	0.058	0.237	0.38
TWD-RCPM	0.026	0.015	0.065	0.53

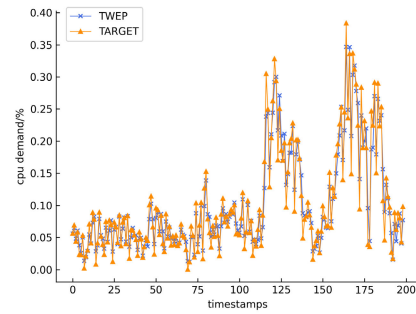
**FIGURE 11.** Task_usage_499 comparison of joint prediction results.**TABLE 4.** Task_usage_499 Comparison of TWD-JPM and TWD-RCPM evaluation indicators.

	RMSE	MAE	SMAPE	QR
TWD-JPM	0.038	0.021	0.101	0.52
TWD-RCPM	0.026	0.015	0.065	0.53

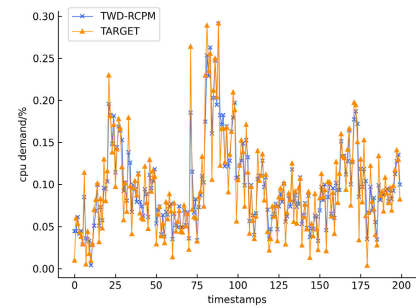
value than TWD-RCPM. The MAE of ARIMA, NN, and DMASVR-3WD are 0.051, 0.05, and 0.058 which is of higher value than TWD-RCPM by 0.036, 0.035, and 0.043, respectively. The SMAPE of ARIMA, NN and DMASVR-3WD are 0.214, 0.215, and 0.237 which is of higher value than TWD-RCPM by 0.149, 0.15, and 0.172, respectively. The QR of ARIMA, NN and DMASVR-3WD are 0.55, 0.48, and 0.38 which is of lower value than TWD-RCPM by 0.02, 0.05, and 0.15, respectively. Based on the above results, NN and DMASVR-3WD have lower QR, and the predicted value is less than the actual value most of the time, which will result in lower resource utilization. Compared with ARIMA, NN, and DMASVR-3WD, the TWD-RCPM reduces the platform cumulative relative predicting error by 69.05%, 68.67%, and 72.63%, respectively, which shows that it has high prediction accuracy and good data fitting capabilities to behave an ideal result by modeling the CPU load characteristics.

In order to verify the effectiveness of the three-way decision, we compared TWD-JPM without error correction and TWD-RCPM. The comparison of the prediction results is shown in Figure 11. The evaluation index calculation results are shown in Table 4.

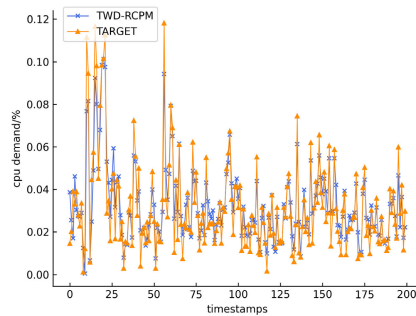
In Figure 11, both TWD-JPM and TWD-RCPM models can effectively improve data fitting ability. Combined with Table 4, there is an over-prediction phenomenon at a few points such as point 7 and point 10, but the overall accuracy and the QR increase. Compared with ARIMA, NN, and DMASVR-3WD, the TWD-JPM model can effectively



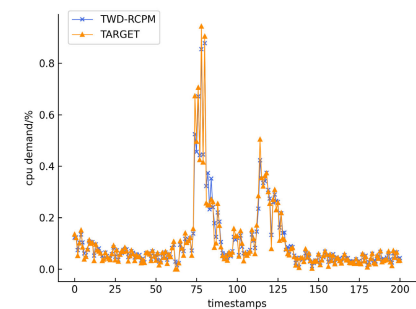
(a) Task_usage_481 prediction result



(b) Task_usage_486 prediction result



(c) Task_usage_495 prediction result



(d) Task_usage_498 prediction result

FIGURE 12. Prediction results of TWD-RCPM model under different tables.

improve the prediction accuracy and the QR can prove the effectiveness of the three-way decision (Table 4 and Table 5).

TABLE 5. Comparison of evaluation indexes of TWD-RCPM prediction model under different data table.

	RMSE	MAE	SMAPE	QR
Task_usage_481	0.047	0.032	0.26	0.54
Task_usage_486	0.053	0.037	0.223	0.51
Task_usage_495	0.025	0.02	0.324	0.59
Task_usage_498	0.064	0.029	0.025	0.57

We choose four Task_usage tables randomly to verify the generalization ability of TWD-RCPM. The prediction results are shown in Figure 12, and the evaluation index calculation results are shown in Table 5.

In Table 5, TWD-RCPM model have a high accuracy compared with other models, and based on values of QR under different data tables, TWD-RCPM model could be proved to have good stability and generalization ability.

V. CONCLUSION

Workload prediction plays an important role in the resource management of data centers. In heterogeneous data centers, it is difficult to predict accuracy due to workload fluctuation. In order to improve the accuracy of the prediction, we proposed a three-way ensemble prediction (TWD-RCPM) for workload in the data center. Based on CPU load monitoring logs from Google data sets, we found that TWD-RCPM can improve the accuracy of workload prediction by 69.05%, 68.67% and 72.6% compared with DMASVR-3WD, ARIMA and NN, respectively, which proved that this approach could predict more accurately.

Our model has generality and scalability, and it could be used in more scenarios, future work includes the extension of our model. The work may include analyzing the factors affecting data fluctuation, in this paper, only CPU modeling is used to build the prediction model. However, the prediction result strongly depends on the type of data set, different resources have different sensitivity to energy consumption. Therefore, we will consider more factors, such as memory, I/O, and disk utilization and introduce more workload prediction models for expanding the system. Besides, further investigation will focus on energy consumptions. The prediction model needs to be applied to data center resource allocation to evaluate the actual energy saving.

REFERENCES

- [1] L. Ismail and H. Materwala, "Computing server power modeling in a data center: Survey, taxonomy, and performance evaluation," *ACM Comput. Surv.*, vol. 53, no. 3, pp. 1–34, May 2020.
- [2] M. A. Elaziz and I. Attiya, "An improved Henry gas solubility optimization algorithm for task scheduling in cloud computing," *Artif. Intell. Rev.*, vol. 54, no. 5, pp. 3599–3637, Jun. 2020.
- [3] B. Yan, F. Hao, and X. Meng, "When artificial intelligence meets building energy efficiency, a review focusing on zero energy building," *Artif. Intell. Rev.*, vol. 54, no. 3, pp. 2193–2220, Mar. 2020.
- [4] E. Antolak and A. Pulka, "Energy-efficient task scheduling in design of multithread time predictable real-time systems," *IEEE Access*, vol. 9, pp. 121111–121127, 2021.
- [5] M. Dayarathna, Y. Wen, and R. Fan, "Data center energy consumption modeling: A survey," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 732–794, 1st Quart., 2017.
- [6] F. F. Zhang, J. Wu, and Z. H. Lv, "Survey on prediction models in cloud resource management schemes," *Comput. Eng. Des.*, vol. 34, no. 9, pp. 3078–3083, 2013.
- [7] N. K. Walia, N. Kaur, M. Alowaidi, K. S. Bhatia, S. Mishra, N. K. Sharma, S. K. Sharma, and H. Kaur, "An energy-efficient hybrid scheduling algorithm for task scheduling in the cloud computing environments," *IEEE Access*, vol. 9, pp. 117325–117337, 2021.
- [8] J. Kumar and A. K. Singh, "Performance evaluation of metaheuristics algorithms for workload prediction in cloud environment," *Appl. Soft Comput.*, vol. 113, Dec. 2021, Art. no. 107895.
- [9] G. Chen, X. Y. Bai, X. F. Huang, M. Y. Li, and L. Z. Zhou, "Cloud performance trend prediction by moving averages," *J. Frontiers Comput. Sci. Technol.*, vol. 6, no. 6, pp. 495–503, 2012.
- [10] R. N. Calheiros, E. Masoumi, R. Ranjan, and R. Buyya, "Workload prediction using ARIMA model and its impact on cloud Applications' QoS," *IEEE Trans. Cloud Comput.*, vol. 3, no. 4, pp. 449–458, Oct. 2015.
- [11] N. I. Sapankevych and R. Sankar, "Time series prediction using support vector machines: A survey," *IEEE Comput. Intell. Mag.*, vol. 4, no. 2, pp. 24–38, May 2009.
- [12] Y. C. Chang, R. S. Chang, and F. W. Chuang, "A predictive method for workload forecasting in the cloud environment," *Lect. Notes Elect. Eng.*, vol. 260, pp. 577–585, 2014.
- [13] H. M. Nguyen, G. Kalra, and D. Kim, "Host load prediction in cloud computing using long short-term memory encoder-decoder," *J. Supercomput.*, vol. 75, no. 11, pp. 7592–7605, Nov. 2019.
- [14] C. Liu, C. Liu, Y. Shang, S. Chen, B. Cheng, and J. Chen, "An adaptive prediction approach based on workload pattern discrimination in the cloud," *J. Netw. Comput. Appl.*, vol. 80, pp. 35–44, Feb. 2017.
- [15] Q. Sun, Z. Tan, and X. Zhou, "Workload prediction of cloud computing based on SVM and BP neural networks," *J. Intell. Fuzzy Syst.*, vol. 39, no. 3, pp. 2861–2867, Oct. 2020.
- [16] J. Patel, V. Jindal, I. L. Yen, F. Bastani, and P. Garraghan, "Workload estimation for improving resource management decisions in the cloud," in *Proc. IEEE 12th Int. Symp. Auton. Decent. Syst. (ISADS)*, Mar. 2015, pp. 25–32.
- [17] C. N. Babu and B. E. Reddy, "A moving-average filter based hybrid ARIMA-ANN model for forecasting time series data," *Appl. Soft Comput.*, vol. 23, pp. 27–38, Oct. 2014.
- [18] M. C. Calzarossa and D. Tessera, "Modeling and predicting temporal patterns of web content changes," *J. Netw. Comput. Appl.*, vol. 56, pp. 115–123, Oct. 2015.
- [19] W. Ran, Z. Yuchao, W. Wendong, X. Ke, and C. Laizhong, "Algorithm of mixed traffic scheduling among data centers based on prediction," *J. Comput. Res. Develop.*, vol. 58, no. 6, pp. 1307–1317, 2021.
- [20] G. Sun, Z. Zhang, B. Zheng, and Y. Li, "Multi-sensor data fusion algorithm based on trust degree and improved genetics," *Sensors*, vol. 19, no. 9, p. 2139, May 2019.
- [21] J. Prassanna and N. Venkataraman, "Adaptive regressive holt-winters workload prediction and firefly optimized lottery scheduling for load balancing in cloud," *Wireless Netw.*, vol. 27, no. 8, pp. 5597–5615, Nov. 2019.
- [22] N. Tran, T. Nguyen, B. M. Nguyen, and G. Nguyen, "A multivariate fuzzy time series resource forecast model for clouds using LSTM and data correlation analysis," *Proc. Comput. Sci.*, vol. 126, pp. 636–645, Jan. 2018.
- [23] Z. Zheng, W. Chen, X. Wu, P. Chen, and J. Liu, "LSTM network: A deep learning approach for short-term traffic forecast," *IET Intell. Transp. Syst.*, vol. 11, no. 2, pp. 68–75, 2017.
- [24] G. V. Houdt, C. Mosquera, and G. Nápoles, "A review on the long short-term memory model," *Artif. Intell. Rev.*, vol. 53, no. 8, pp. 5929–5955, 2020.
- [25] Y. Yao, "Three-way decisions with probabilistic rough sets," *Inf. Sci.*, vol. 180, no. 3, pp. 341–353, Feb. 2010.
- [26] C. Jiang, D. Guo, and R. Xu, "Measuring the outcome of movement-based three-way decision using proportional utility functions," *Int. J. Speech Technol.*, vol. 51, no. 12, pp. 8598–8612, Dec. 2021.
- [27] C. Jiang and Y. Yao, "Effectiveness measures in movement-based three-way decisions," *Knowl.-Based Syst.*, vol. 160, pp. 136–143, Nov. 2018.
- [28] C. M. Jiang, J. W. Wu, and Z. C. Li, "Adaptive thresholds determination for saving cloud energy using three-way decisions," *Cluster Comput.*, vol. 22, pp. 8475–8482, Feb. 2019.

- [29] X. X. Xu, C. M. Jiang, and C. M. Huang, "A method of energy-aware mobile cloud tasks offloading based on three-way decision," *J. Nanjing Univ. Sci. Technol.*, vol. 43, no. 4, pp. 447–454, 2019.
- [30] S. Luo, "Three-way decision in a multi-source information system and its applications," *IEEE Access*, vol. 7, pp. 108343–108359, 2019.
- [31] C. Jiang, D. Guo, Y. Duan, and Y. Liu, "Strategy selection under entropy measures in movement-based three-way decision," *Int. J. Approx. Reasoning*, vol. 119, pp. 280–291, Apr. 2020.
- [32] C. Jiang, L. Yang, and R. Shi, "An energy-aware virtual machine migration strategy based on three-way decisions," *Energy Rep.*, vol. 7, pp. 8597–8607, Nov. 2021.
- [33] J. Peng, J. Chen, X. Zhi, M. Qiu, and X. Xie, "Research on application classification method in cloud computing environment," *J. Supercomput.*, vol. 73, no. 8, pp. 3488–3507, Aug. 2017.
- [34] X. Zhang, Q. Zhang, Y. Cheng, and G. Wang, "Optimal scale selection by integrating uncertainty and cost-sensitive learning in multi-scale decision tables," *Int. J. Mach. Learn. Cybern.*, vol. 11, no. 5, pp. 1095–1114, May 2020.
- [35] X. Y. Jia and L. Shang, "A simulated annealing algorithm for learning thresholds in three-way decision-theoretic rough set model," *J. Chin. Comput. Syst.*, vol. 34, no. 11, pp. 2603–2606, 2013.
- [36] M. Abdel-Basset, K. M. Sallam, R. Mohamed, I. Elgendi, K. Munasinghe, and O. M. Elkomy, "An improved binary grey-wolf optimizer with simulated annealing for feature selection," *IEEE Access*, vol. 9, pp. 139792–139822, 2021.
- [37] T. Tlili and S. Krichen, "A simulated annealing-based recommender system for solving the tourist trip design problem," *Expert Syst. Appl.*, vol. 186, Dec. 2021, Art. no. 115723.
- [38] Z. Shen, H. Yang, and S. Zhang, "Deep network with approximation error being reciprocal of width to power of square root of depth," *Neural Comput.*, vol. 33, no. 5, pp. 1–32, 2021.
- [39] M. Khadem-Hosseini, S. Ghaemmaghami, A. Abtahi, S. Gazor, and F. Marvasti, "Error correction in pitch detection using a deep learning based classification," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 990–999, 2020.
- [40] R. F. da Silva, H. Casanova, A.-C. Orgerie, R. Tanaka, E. Deelman, and F. Suter, "Characterizing, modeling, and accurately simulating power and energy consumption of I/O-intensive scientific workflows," *J. Comput. Sci.*, vol. 44, Jul. 2020, Art. no. 101157.
- [41] C. Reiss, A. Tumanov, G. R. Ganger, R. H. Katz, and M. A. Kozuch, "Heterogeneity and dynamicity of clouds at scale: Google trace analysis," in *Proc. 3rd ACM Symp. Cloud Comput. - SoCC*, 2012.



RUI SHI was born in Daxing'anling, Heilongjiang, China. She is currently pursuing the master's degree with Harbin Normal University. Her main research interests include three-way decision and cloud computing.



CHUNMAO JIANG received the M.S. degree from the Harbin Institute of Technology, Harbin, Heilongjiang, China, in 2004, and the Ph.D. degree in computer science from Harbin Engineering University, Harbin. He is currently a Professor at Harbin Normal University. His research interests include cloud computing and embedded computing.

...